Topic 1: Course Overview and Introduction

Course Information

- CPR E 4890: Computer Networking and Data Communications
  - Lecture Time: TR 11:00 AM ~ 12:15 PM
  - Location: 1312 Hoover
  - Prerequisite: CPR E 2880 or COM S 3270
  - Course Homepage: Canvas
- Instructor:
  - Daji Qiao (daji@iastate.edu), Professor
  - Office: 313 Durham Center
  - Tel: (515) 294 2390
  - Office Hour: W 12-1 @ 313 Durham and/or on Webex
- Teaching Assistants:
  - Joseph Schmidt (jschm333@iastate.edu)
    - Office Hour: F 1-2 @ 2061 Coover and/or on Webex
  - Shahab Afshar (safshar@iastate.edu)
    - Office Hour: M 2-3 @ 2061 Coover and/or on Webex

Textbook Information

- No required textbook

- Recommended books:
  - A. Leon-Garcia and I. Widjaja, *Communication Networks: Fundamental Concepts and Key Architectures*, 2^{nd} Edition, McGraw-Hill, 2004.
  - J.F. Kurose and K.W. Ross, *Computer Networking: A Top-Down Approach,* 6^{th} Edition, Pearson, 2012
  - W.R. Stevens, B. Fenner, and A.M. Rudoff, *Unix Network Programming, Volume 1: The Sockets Networking API*, 3^{rd} Edition, Addison-Wesley, 2003

Tentative Lecture Coverage

Topics	**Coverage (# lectures)**
Introduction to Computer Networking	**2**
Physical Layer • Digital Transmission Fundamentals • Line Coding	**2**
Error Detection and Recovery • Basic Error Detection Codes • CRC (Cyclic Redundancy Check) • Retransmission Strategies (ARQ Protocols)	**8**
Data Link Layer • Random Access • MAC (Medium Access Control) • LAN (Local Area Network) • Ethernet • Etc.	**4**

Tentative Lecture Coverage

Topics	Coverage (# lectures)
Network Layer • IP Addresses • ARP, RARP, DHCP, NAT, ICMP • Routing Protocols • Etc.	**6**
Transport Layer • TCP Protocol • TCP Error Control • TCP Flow Control • TCP Congestion Control • Etc.	**6**
Midterm Exam	**1**
Review & Recitation	**1**
	Total: 30 (15 weeks)

Homework Assignments

- Homework Assignments
  - We will have up to 6 homework assignments.
  - Each student will work on HWs individually.
  - HWs are due one week from the assign date, unless specified otherwise.
  - Please make sure to check Canvas about the exact due date and time.

Lab Assignments

- Lab Assignments
  - We will have up to 10 lab assignments.
  - Each student will work on labs individually, unless specified otherwise.
  - Lab attendance is required.
  - Lab location: 2061 Coover
    - Section A: Tue 2:15 – 4:05 PM (TA: Joe)
    - Section B: Tue 4:25 – 6:10 PM (TA: Joe)
    - Section C: Wed 1:10 – 3:05 PM (TA: Shahab)
  - Lab reports are due one week from the assign date, unless specified otherwise.
  - Please make sure to check Canvas about the exact due date and time.

Week	Lab Information (Tentative)
1	First Week
2	Lab #1: Network Utility Programs
3	Lab #2: TCP Sockets Programming
4	Lab #3: UDP Sockets Programming
5	Lab #4: CloudLab – Basics
6	Lab #5: Error Recovery with Go-Back-N ARQ Protocol
7	
8	Midterm
9	Spring Break
10	
11	Lab #6: CloudLab – Static Routing
12	Lab #7: Using Cisco IOS to Configure Cisco Routers
13	Lab #8: Using Cisco IOS to Configure OSPF Routing
14	Lab #9: CloudLab – TCP Congestion Control
15	Lab #10: Advanced Topic
16	Prep Week

Late Submission Policy

- For all assignments, the following Late Submission Policy will be adopted:
  - 10% per day penalty for the first 5 weekdays (weekend days are exempted).
  - For example, for an assignment due Monday:
    - Tuesday submission carries 10% penalty
    - Wednesday submission carries 20% penalty
    - Thursday submission carries 30% penalty
    - Friday, Saturday, Sunday submissions carry 40% penalty
    - (Next) Monday submission carries 50% penalty
  - Any submission after one week receives zero credit.

Exam Information

- All quizzes and exams are open books/notes/references/assignments.

- Midterm exam:
  - 03/12 (Thu) 11:00 AM ~ 12:15 PM @ 1312 Hoover

- Final exam:
  - 05/13 (Wed) 9:45 ~ 11:45 AM @ 1312 Hoover
  - Final exam is comprehensive.

Grading Information

	Percentage	Per Assignment
Homework Assignments (~6)	15%	~2.5%
Lab Assignments (~10)	25%	~2.5%
Quizzes	5%	~1%
Midterm Exam	25%	25%
Final Exam	30%	30%
	Total: 100%	

Course Grade	A	A-	B+	B	B-	C+	C	C-	D
Cumulative Total	90+	87+	83+	80+	77+	73+	70+	60+	50+

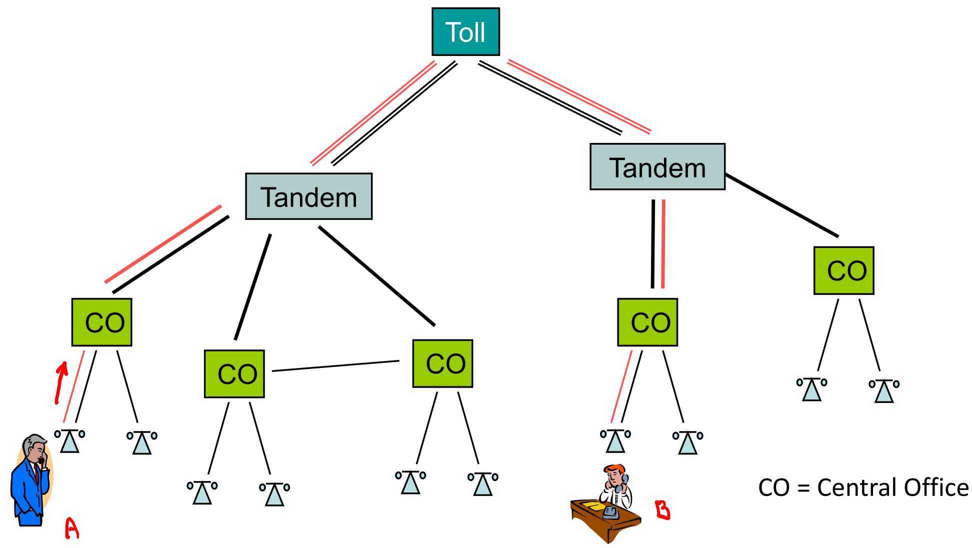
Internet is a **packet-switching** communication network.

Network Architecture and Transfer Mode

- Communication Network
  - A set of equipment and facilities that enable the transfer of information between users located at different locations
- Network Architecture
  - Specifies how the network is built and operated
  - Specifies how information is transferred (transfer mode)
- Three Network Architectures and Transfer Modes
  - Telephone network – circuit switching
  - Telegraph network – message switching
  - Computer network – packet switching

Circuit Switching in Telephone Networks

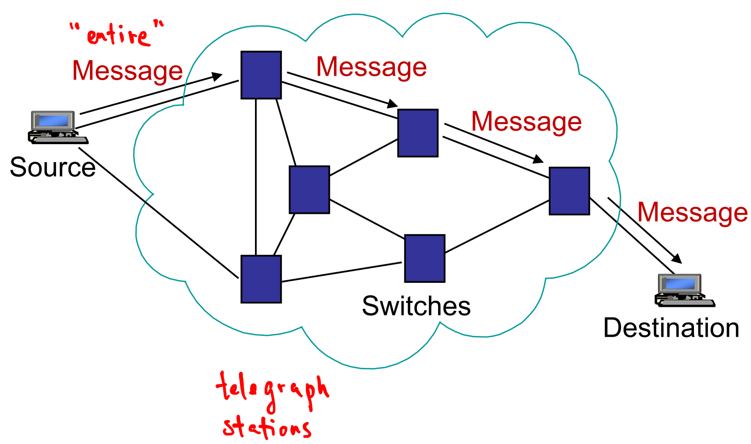
- Circuit Switching – *"Reserve and Use"*
  - Automated switches set up a physical circuit between two ends
  - All messages follow the same path (via the established circuit)



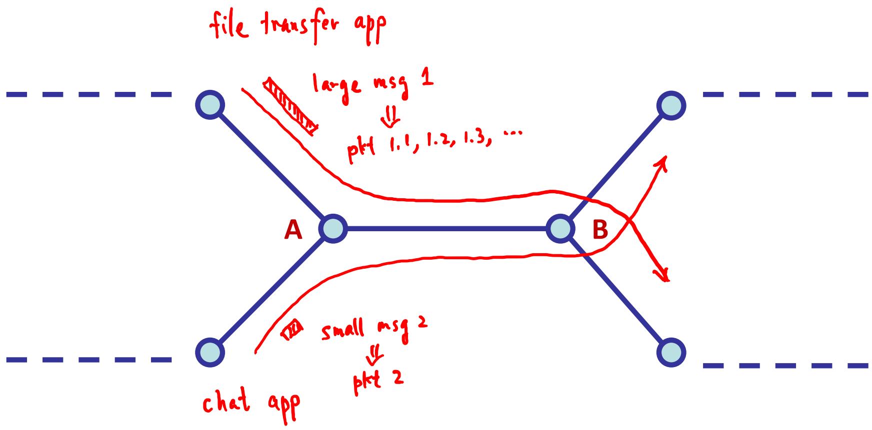
Message Switching in Telegraph Networks

- Message Switching – *"Store and Forward"*
  - Store-and-Forward Operation
  - Addressing, Routing, Forwarding

Telegram
= content + control info



Message Switching for Computer Networks?



Message Switching : "entire" message
- not fair
- not flexible
- cannot provide desired QoS (Quality of Service)

Packet Switching in Computer Networks

- Packet Switching – *"Break and Route"*
  - Break long messages into packets
  - Packets have maximum length
  - Network transfers packets using store-and-forward

  - Requires: Addressing, Framing, Routing, etc.
  - Intelligence is at the edge of the network